Probability & Statistics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Random sampling**

What is the value of $16^{\frac{3}{4}}$?

By the definition of rational exponents, $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$ for any positive real number a. Here, $a = 16$, $m = 3$, and $n = 4$: $16^{\frac{3}{4}} = (\sqrt[4]{16})^3$ Since $2^4 = 16$, the principal fourth root is $\sqrt[4]{16} = 2$. Raising this result to the power of 3 gives: $2^3 = 8$ Common misconceptions: - Multiplying the base by the fraction gives $16 \cdot \frac{3}{4} = 12$, which incorrectly treats the exponent as a multiplier. - Finding $\sqrt[4]{16} = 2$ and then multiplying by 3.

2 **[Foundational]** • **Target: Estimating proportions from samples**

Which expression is equivalent to $(16x^8y^4)^{\frac{3}{4}}$ for all positive values of x and y?

To simplify the expression $(16x^8y^4)^{\frac{3}{4}}$, apply the power of a product rule $(abc)^n = a^n b^n c^n$ and the power of a power rule $(a^m)^n = a^{mn}$: 1. Evaluate the numerical coefficient: $16^{\frac{3}{4}} = \left(\sqrt[4]{16}\right)^3 = 2^3 = 8$ 2. Apply the exponent to the variable x: $(x^8)^{\frac{3}{4}} = x^{8 \cdot \frac{3}{4}} = x^{\frac{24}{4}} = x^6$ 3. Apply the exponent to the variable y.

3 **[Foundational]** • **Target: Describing shapes of data distributions**

Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. What is the value of $(f \circ g)(2)$?

To find the composite function value $(f \circ g)(2)$, we evaluate the inner function first, and then apply the outer function: 1. Evaluate $g(2)$: $g(2) = 3(2) - 2 = 6 - 2 = 4$ 2. Substitute this result into $f(x)$: $(f \circ g)(2) = f(g(2)) = f(4) = 4^2 + 1 = 16 + 1 = 17$ Common misconceptions: - Reversing the order of composition to compute $(g \circ f)(2) = g(f(2)) = g(2^2 + 1) = g(5) = 3(5) - 2 = 13$ yields choice 13. - Mistaking composition for multiplication.

4 **[Foundational]** • **Target: The normal distribution**

What are the solutions to the quadratic equation $x^2 - 6x + 13 = 0$?

To solve the quadratic equation $x^2 - 6x + 13 = 0$, use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ with $a = 1$, $b = -6$, and $c = 13$: $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(13)}}{2(1)} = \frac{6 \pm \sqrt{36 - 52}}{2} = \frac{6 \pm \sqrt{-16}}{2}$ Since $\sqrt{-16} = 4i$, simplify the expression: $x = \frac{6 \pm 4i}{2} = 3 \pm 2i$ - $x = 3 \pm 4i$ occurs if the radical term is not divided by the denominator 2. - $x = -3 \pm 2i$ occurs from a sign error when computing $-b$..

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5 [Foundational] • Target: Interpreting and comparing z-scores

What are the center and radius of the circle given by the equation $(x-2)^2+(y+6)^2=36$?

The standard form for the equation of a circle is: $(x-h)^2+(y-k)^2=r^2$ where (h,k) is the center of the circle and r is the radius. Comparing $(x-2)^2+(y+6)^2=36$ to standard form: 1. $x-h=x-2\Rightarrow h=2$ 2. $y-k=y-(-6)\Rightarrow k=-6$ 3. $r^2=36\Rightarrow r=\sqrt{36}=6$ Thus, the center is $(2,-6)$ and the radius is 6. Common errors: - Reversing the signs of the coordinates leads to $(-2,6)$. - Forgetting to take the square root of the constant term leads t.

6 [Foundational] • Target: Normal distribution probabilities

A sequence is defined by the recursive formula $a_n=2a_{n-1}+1$, with the first term $a_1=3$. What is the value of the third term, a_3?

The recursive formula for the sequence is $a_n=2a_{n-1}+1$, and the first term is $a_1=3$. To find the second term, a_2, we substitute $n=2$ into the formula: $a_2=2a_{2-1}+1=2a_1+1$ Since $a_1=3$, we have: $a_2=2(3)+1=6+1=7$ To find the third term, a_3, we substitute $n=3$ into the formula: $a_3=2a_{3-1}+1=2a_2+1$ Since $a_2=7$, we have: $a_3=2(7)+1=14+1=15$

7 [Foundational] • Target: Random sampling

Consider the quadratic equation $x^2-8x+15=0$. What are the roots of this equation?

To find the roots of the quadratic equation $x^2-8x+15=0$, we can factor the quadratic expression. We are looking for two numbers that multiply to 15 and add up to -8. These numbers are -3 and -5. Therefore, the factored form of the equation is $(x-3)(x-5)=0$. Setting each factor to zero, we get $x-3=0$ which gives $x=3$, and $x-5=0$ which gives $x=5$. Thus, the roots are $x=3$ and $x=5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Estimating proportions from samples

Consider the matrix equation: $(3,-1;2,k)(x;y)=(5;7)$. If the matrix $(3,-1;2,k)$ is singular, what is the value of k?

A matrix is singular if and only if its determinant is zero. The determinant of a 2x2 matrix $(a,b;c,d)$ is given by $ad-bc$. For the given matrix $(3,-1;2,k)$, the determinant is $(3)(k)-(-1)(2)=3k+2$. For the matrix to be singular, the determinant must be zero: $3k+2=0$. Solving for k: $3k=-2$, which gives $k=-2/3$.

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9 [Intermediate] • Target: Describing shapes of data distributions

What is the center of the circle represented by the equation $(x-3)^2+(y+5)^2=16$?

The standard form of a circle's equation is $(x-h)^2+(y-k)^2=r^2$, where (h, k) is the center of the circle. In the given equation, $(x-3)^2+(y+5)^2=16$, we can identify $h=3$ and $k=-5$. Therefore, the center of the circle is $(3, -5)$.

10 [Intermediate] • Target: The normal distribution

Solve: $\sqrt{2x+5}-\sqrt{x+2}=1$.

Isolating $\sqrt{2x+5}=1+\sqrt{x+2}$ and squaring gives $2x+5=x+3+2\sqrt{x+2} \Rightarrow x+2=2\sqrt{x+2}$. Since $\sqrt{x+2}=2$ (as $\sqrt{x+2}=0$ gives $x=-2$, which is extraneous), we get $x+2=4 \Rightarrow x=2$.

11 [Intermediate] • Target: Describing shapes of data distributions

For the model $f(x)=3x+4$ in describing shapes of data distributions, evaluate $f(5)$.

Final Answer: 19. Substitute $x=5$: $f(5)=3(5)+4=19$.

12 [Intermediate] • Target: Describing shapes of data distributions

Find the positive solutions to the quadratic equation $x^2-13x+36=0$.

Final Answer: $x=4$ and $x=9$. Factoring $(x-4)(x-9)=0$ yields the roots $x=4$ and $x=9$.

13 [Intermediate] • Target: The normal distribution

For the model $f(x)=4x+5$ in the normal distribution, evaluate $f(2)$.

Final Answer: 13. Substitute $x=2$: $f(2)=4(2)+5=13$.

14 [Intermediate] • Target: The normal distribution

Find the positive solutions to the quadratic equation $x^2-4x+3=0$.

Final Answer: $x=1$ and $x=3$. Factoring $(x-1)(x-3)=0$ yields the roots $x=1$ and $x=3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The normal distribution • Cross-Lesson: 6.01 Radicals & Rational Exponents

A data set recorded for the normal distribution consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10+14+18+24=66$, and dividing by 4 gives 16.5.

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16 [Challenging] • Target: The normal distribution • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(3)$.

Final Answer: 17. Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

17 [Challenging] • Target: The normal distribution • Cross-Lesson: 8.01 Conic Sections

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

Final Answer: $x = 2$ and $x = 4$. Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

18 [Challenging] • Target: The normal distribution • Cross-Lesson: 9.01 Trigonometric Functions

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(4)$.

Final Answer: 21. Substitute $x = 4$: $f(4) = 4(4) + 5 = 21$.

19 [Challenging] • Target: The normal distribution • Cross-Lesson: 1.01 Functions & Their Graphs

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

Final Answer: $x = 3$ and $x = 5$. Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

20 [Challenging] • Target: The normal distribution • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(5)$.

Final Answer: 25. Substitute $x = 5$: $f(5) = 4(5) + 5 = 25$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Random sampling

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(2)$.

a. 21 ✓

b. 27

c. 12

d. 25

Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

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22 [Foundational] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

23 [Foundational] • Target: Random sampling

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(5)$.

a. 39 ✓

b. 45

c. 30

d. 43

Substitute $x = 5$: $f(5) = 6(5) + 9 = 39$.

24 [Foundational] • Target: Estimating proportions from samples

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

25 [Foundational] • Target: Estimating proportions from samples

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

26 [Foundational] • Target: Estimating proportions from samples

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(5)$.

a. 13 ✓

b. 15

c. 10

d. 17

Substitute $x = 5$: $f(5) = 2(5) + 3 = 13$.

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27 [Foundational] • Target: Describing shapes of data distributions

For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

28 [Foundational] • Target: Describing shapes of data distributions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

30 [Intermediate] • Target: Random sampling

A data set recorded for random sampling consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

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31 [Intermediate] • Target: Random sampling

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(3)$.

a. 27 ✓

b. 33

c. 18

d. 31

Substitute $x = 3$: $f(3) = 6(3) + 9 = 27$.

32 [Intermediate] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

33 [Intermediate] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

34 [Intermediate] • Target: Estimating proportions from samples

A data set recorded for estimating proportions from samples consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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35 [Intermediate] • Target: Estimating proportions from samples

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(3)$.

a. 9 ✓

b. 11

c. 6

d. 13

Substitute $x = 3$: $f(3) = 2(3) + 3 = 9$.

36 [Intermediate] • Target: Estimating proportions from samples

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

37 [Intermediate] • Target: Estimating proportions from samples

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Describing shapes of data distributions • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for describing shapes of data distributions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

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39 [Challenging] • Target: Describing shapes of data distributions • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

40 [Challenging] • Target: Describing shapes of data distributions • Cross-Lesson: 10.01 Matrices & Systems

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

41 [Challenging] • Target: The normal distribution • Cross-Lesson: 2.01 Quadratics & Complex Numbers

A data set recorded for the normal distribution consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

42 [Challenging] • Target: Interpreting and comparing z-scores • Cross-Lesson: 3.01 Polynomial Functions

A data set recorded for interpreting and comparing z-scores consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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43 [Challenging] • Target: Interpreting and comparing z-scores • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 5x + 6$ in interpreting and comparing z-scores, evaluate $f(3)$.

a. 21 ✓

b. 26

c. 15

d. 25

Substitute $x = 3$: $f(3) = 5(3) + 6 = 21$.

44 [Challenging] • Target: Normal distribution probabilities • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A data set recorded for normal distribution probabilities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

45 [Challenging] • Target: Normal distribution probabilities • Cross-Lesson: 6.01 Radicals & Rational Exponents

For the model $f(x) = 6x + 7$ in normal distribution probabilities, evaluate $f(3)$.

a. 25 ✓

b. 31

c. 18

d. 29

Substitute $x = 3$: $f(3) = 6(3) + 7 = 25$.

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